THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 2

Chapter 8: **Heredity** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. If a human sperm has 22 autosomes and the Y chromosome, the number of chromosomes is 1

(a) 44 **(b)** 22 **(c)** 23 **(d)** 46

2. A short plant sprayed with a growth hormone becomes tall. Its offspring will be 1

(a) short **(b)** dwarf only in F1 **(c)** tall **(d)** medium

3. A plant with genotype RrYy forms how many types of gametes? 1

(a) 16 **(b)** 8 **(c)** 2 **(d)** 4

4. In a human egg, the number of autosomes is 1

(a) 23 **(b)** 22 **(c)** 46 **(d)** 44

5. The expected genotypic ratio of the cross Tt x Tt is 1

(a) 1 : 2 : 1 **(b)** 9 : 3 : 3 : 1 **(c)** 1 : 1 **(d)** 3 : 1

6. In a dihybrid cross, if 144 F2 seeds are obtained, expected round green seeds are 1

(a) 9 **(b)** 81 **(c)** 54 **(d)** 27

7. Total autosomes in a normal human zygote 1

(a) 22 **(b)** 23 **(c)** 44 **(d)** 46

8. The probability that offspring of Tt x Tt are heterozygous is 1

(a) 1 **(b)** 1/2 **(c)** 1/4 **(d)** 3/4

9. In a dihybrid F2, the proportion of seeds with both recessive traits is 1

(a) 9/16 **(b)** 3/16 **(c)** 1/16 **(d)** 1/4

10. Which cell has half the number of chromosomes of a body cell? 1

(a) Skin cell **(b)** Muscle cell **(c)** Zygote **(d)** Sperm

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. How does a dihybrid cross show that two traits are inherited independently? 2

12. How can two parents with free earlobes have a child with attached earlobes? 2

13. Why is it impossible for a mother to pass a Y chromosome to her child? 2

14. A woman has three daughters and wishes for a son. Explain why her genes cannot be responsible. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. In pea plants, yellow (Y) is dominant to green (y). A yellow plant crossed with a green plant gives 50 yellow and 48 green plants. Find the genotype of the yellow parent. Give the reasoning. 3

16. Why is the proportion of boys and girls expected to be nearly equal in a large population? 3

17. How is a trait expressed from gene to protein? Explain using the example of plant height. 3

OR

Explain why some traits are inherited and some are not, giving a suitable example for each. 3

18. Differentiate between the monohybrid and dihybrid crosses on the basis of number of characters, F1 and F2 ratios. 3

OR

Describe the contribution of chromosomes of each parent to a child's genetic make-up. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain with the help of a cross how a Tt tall plant crossed with a tt plant helps to identify the genotype of an unknown tall plant. 5

OR

Explain how variation is inherited and accumulated. Compare the amount of variation in an asexually reproducing organism with that in a sexually reproducing organism. 5

20. Explain how the sex of the child is determined and why the number of boys and girls is almost equal. Is the mother responsible for the birth of a girl? 5

OR

Describe how the chromosome number is maintained from parents to offspring, and how the sex chromosomes of both parents decide the sex. 5

END OF QUESTION PAPER — Total: 40 Marks

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